GEOAI HACKATHON @ CSRE, IIT BOMBAY · 2026

GeoAI Outdoor Planner

Where and when to head outdoors, for this person, reasoned from live map, weather and air-quality data.

Small open model: Qwen3.5-2B + LoRA, run locally



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THE PROBLEM AND THE APPROACH

The right answer depends on the hour and the place

A ground downwind of an expressway at 8 AM can be pleasant by evening once the wind shifts. A venue 40 minutes away at rush hour may not be worth it. And the limits change with the person: asthma, a toddler, a pregnancy.

Live data, every question

OpenStreetMap, Open-Meteo, CAMS air quality and OSRM, fetched for each question.

Code does the numbers

Every candidate place is scored for every hour, with rules you can read.

The small model does language

It reads messy questions and writes the answer from a checked fact sheet.

How a question becomes a grounded answer

Blue steps: small language model · **Orange steps:** deterministic code

1 Understand

The SLM turns the question into JSON: activity, place, time, radius, profile, question type.

2 Find places

Geocode with Nominatim; venues from OpenStreetMap via Overpass, or the offline Geofabrik index.

3 Fetch live data

Open-Meteo hourly weather, CAMS air quality, OSRM travel; nearby roads, industry, water, greenery.

4 Score place × hour

Activity limits tightened by profile; downwind of roads and industry each hour; daylight; traffic.

5 Write the answer

The SLM writes from a fact sheet that carries the computed decision, and nothing else.

6 Check, then return

Numbers, places, days, AQI labels and the decision must match the sheet, or a fixed writer answers.

WORKED EXAMPLE · LIVE RUN

“Is Saturday or Sunday better for a hike near Lonavala?”

Sunday is the better day (day score 82 vs 76). Best: **Lohagad Fort, Sun 07:00–11:00**.

Place	Distance	Sunday slot	Score
Lohagad Fort	8.6 km · ~13 min	07:00–11:00	86
Morgiri Fort	9.6 km · ~30 min	07:00–11:00	78
Tikona Fort	17.3 km · ~29 min	07:00–11:00	75

Carry: 2+ L water, trail shoes with grip, torch, offline map.

Why, from the data

- Sun 07:00 at Lohagad: feels like 20°C, rain 0%, wind from E 10 km/h, US AQI 107
- Saturday’s best slot: US AQI 127; avoid Sat 11:00–17:00 (air quality)
- Before 07:00 and after 18:00 it’s dark (sunset 18:06), so no starts there

Understood as: hiking · Lonavala · 25 km · Sat and Sun · compare

MODEL AND FINE-TUNING

A 1.9B-parameter model, fine-tuned for two jobs

- Qwen3.5-2B (Apache-2.0), loaded text-only: 1.88B parameters
- LoRA r=16 on attention, linear-attention and MLP layers: 15.6M trainable (0.82%)
- 25 minutes on one RTX 4000 Ada (20 GB)
- Training data built by code, no other LLM: 2,400 questions with known labels; 434 fact sheet → answer pairs from real pipeline runs, with monsoon, heatwave, fog, smog and storm scenarios

Measure	Base, few-shot	+ LoRA
Parse: all key fields correct	71.5%	100%
Answers passing all checks	64.9%	100%
Live test: SLM answers passing, of 24	20	24
Median latency, live test	12.2 s	8.6 s

The first two rows use held-out data from the same generators as training. The live 30-question test is the honest measure.

LIMITATIONS

An honest note on what it can’t do yet

No live traffic

No allowed free source has it: OSRM free-flow times × a rush-hour multiplier.

Downwind is an estimate

Geometry plus forecast wind, not a measurement; capped so it can’t dominate.

Coarse air quality

CAMS is ~40 km resolution and ~5 days ahead; local hotspots are missed.

OSM is uneven

Unnamed, mis-tagged or missing venues; lighting and access tags are often absent.

Overpass overloads

After 12 s it uses a Geofabrik snapshot; each answer says which it used.

Wording, not facts

Checks catch invented numbers, places, days and decisions, not every nuance.